

DESIGN AND IMPLEMENTATION OF AN INTELLIGENT BATTERY MANAGEMENT SYSTEM (AI-PBMS)

A Physics-Informed Machine Learning Framework for Early Predictive Battery Fault Diagnostics

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1. PROBLEM STATEMENT & MOTIVATION

- **The Critical Failure of Conventional Threshold BMS:** Standard commercial BMS units operate strictly on reactive, static cutoffs (e.g., disconnecting only *after* a cell drops below 2.8V or exceeds 55°C). By the time static thresholds are breached, internal degradation, localized dendrite formation, or irreversible thermal escalation is already underway.
- **The Black-Box AI Challenge:** Pure machine learning algorithms without physical constraints suffer from false positives during rapid acceleration/regen pulses and unpredictable hallucinations when operating out-of-distribution.
- **Our Solution (AI-PBMS):** We engineered an edge-deployed, dual-layer Battery Management System combining a **28-feature time-series XGBoost machine learning model** for early trend prediction with an electrochemical physics arbitration engine to guarantee safety, zero false negatives, and sub-second fault isolation.

2. BATTERY PACK SPECIFICATIONS & EDGE HARDWARE PIPELINE

- **Battery Pack Configuration:** 8S2P pack constructed using LG Energy Solution INR21700-M50 cylindrical Li-ion cells (NMC-811 chemistry, 3.63 V nominal, 5000 mAh per cell; Pack nominal: 29.04 V, 10 Ah, ~290 Wh).
- **Operating Boundaries:** Full-charge cutoff: 33.60 V (4.20 V/cell); Discharge cutoff: 22.40 V (2.80 V/cell); Max continuous discharge current: 14.55 A (pulse: 20 A).
- **Industrial BMS Sensing Interface:** An industrial JBD smart BMS board connects directly across all series balance taps and monitors spatial thermal distribution through four surface-mounted NTC thermistor probes.
- **Raspberry Pi Edge Gateway (bms_bluetooth_gateway.py):**
 - Connects over Bluetooth Low Energy (BLE) to the BMS GATT UART profile, continuously streaming binary telemetry frames at 1.0–2.0 Hz.
 - Decodes packet structures (commands 0x03 and 0x04) into 8 individual cell voltages (mV resolution), pack current, Coulomb-counted state-of-charge (SOC), and 4 temperature channels.
 - **Autonomous Edge Resilience:** Appends every packet to local recovery CSV/Excel logs (bms_local_telemetry_log.csv) and forwards token-authenticated JSON frames to the cloud API endpoint.

3. MACHINE LEARNING ARCHITECTURE & 28-FEATURE TIME-SERIES PIPELINE

- **Temporal Sliding Window Inference:** The ML engine ingests a sliding 60-second temporal buffer (60 consecutive rows) to capture multi-rate dynamic battery trends rather than isolated instantaneous points.
- **28 Dynamic Engineered Features:**
 - *Electrochemical First Derivatives:* Temporal rates of change (dV/dt , dI/dt , dT/dt , $dSOC/dt$) to detect anomalous acceleration trajectories.
 - *Rolling Statistical Dynamics:* 10-sample moving standard deviations (σ_V , σ_T) and rolling means to identify noise and impedance volatility.
 - *Spatial Thermal Dispersion:* Multi-point gradient ($\Delta T = \max(NTC_{1..4}) - \min(NTC_{1..4})$) isolating localized hot-spot formation from ambient heat.
 - *Individual Cell Dynamics:* Per-cell differential drop rates ($\Delta V_{cell_i}/\Delta t$) and rise rates under high charge/discharge transients.
- **Out-of-Distribution (OOD) Envelope Guard:** Normalized feature bounds evaluate operational uncertainty. If incoming sensor patterns diverge significantly from trained empirical profiles, the system flags an OOD state to prevent false classifications.
- **Dynamic Driving Mode Classifier:** Analyzes current polarity and magnitude to automatically classify operation into IDLE, ACCELERATION, CRUISING, and REGENERATION/DECELERATION, adjusting feature baselines dynamically.

DESIGN AND IMPLEMENTATION OF AN INTELLIGENT BATTERY MANAGEMENT SYSTEM (AI-PBMS)

Predictive Anomaly Signatures, Closed-Loop Safety Arbitration & Experimental Results

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4. DYNAMIC ML ANOMALY SIGNATURES & PREDICTIVE DETECTION MECHANISM

Rather than waiting for reactive threshold trips, the AI-PBMS analyzes dynamic temporal trajectories to identify faults during the early **Warning/Risk Phase** before physical safety boundaries are breached.

Fault Classification	Dynamic ML Signature & Time-Series Pattern	Electrochemical & Physical Manifestation	Detection Mechanism
Cell Imbalance	Divergence rate $d(\Delta V)/dt > 0$; sliding variance σ_V widening across cycle transitions; non-uniform Coulombic efficiency during CC discharge.	Capacity mismatch across parallel/series groups; premature capacity fade; uneven electrode degradation.	Predictive AI: Gradient-boosted time-series classifier detects divergent trajectory 45s early. Physics Failsafe: Trip at $\Delta V \geq 150$ mV.
Weak Cell (Degradation)	Asymmetric dynamic voltage drop ($\Delta V_{\text{sag}} > 20\%$ vs pack) under high load current pulses; sluggish post-pulse relaxation curve (dV/dt_{relax}).	Electrolyte dry-out, SEI layer growth, or current collector degradation increasing internal resistance (R_i).	Predictive AI: Evaluates transient impedance sag ($R_i \approx \Delta V/\Delta I$) in ACCEL mode before cutoff. Physics Failsafe: $V_{\text{cell}} \leq 2.80$ V.
Overvoltage / Regen Runaway	Hyperbolic dV/dt acceleration near terminal SOC; rapid voltage jump exceeding electrochemical equilibrium under regen braking.	Lithium plating on anode; cathode lattice degradation; electrolyte oxidation causing gas generation.	Predictive AI: Trajectory model projects time-to-overcharge; throttles regen. Physics Failsafe: Trip at 4.25 V / 33.6 V.
Undervoltage / Deep Discharge	Non-linear knee-point voltage decay; sharp negative dV/dt slope; cell voltage collapsing disproportionate to load current.	Copper dissolution through separator; cell polarity reversal; irreversible active material loss.	Predictive AI: Identifies transition into rapid exhaustion knee 60s ahead. Physics Failsafe: Cutoff at 2.80 V.
Overtemperature & Hotspots	Steep positive dT/dt ($>0.1^\circ\text{C/s}$) coupled with spatial dispersion ($\Delta T_{\text{NTC}} > 5^\circ\text{C}$); joule heating outpacing dissipation curves.	Exothermic SEI decomposition; localized micro-short circuit; thermal dissipation bottleneck.	Predictive AI: Covariance model detects core heat rise early. Physics Failsafe: Cutoff at 55°C (Emergency: 60°C).
Sensor Failure / Cell Dropout	Instantaneous step-function collapse to ~ 0 V ($ dV/dt > 2$ V/s) with zero current swing ($dI/dt \approx 0$); non-physical continuity break.	Disconnected balance wire, blown sensing fuse, or sensor ADC rail hardware failure.	True Cell Integrity: Discards zero-padding; isolates channel; triggers immediate CRITICAL alert.

5. PHYSICS-INFORMED SAFETY ARBITRATION & OUTBOUND ALERTING PIPELINE

- Deterministic Safety Override (Zero False Negatives):** While the ML engine provides continuous predictive classification, safety is anchored by deterministic physics arbitration. If the ML model classifies a state as 'Normal' but Layer 1 physical bounds are violated (e.g., cell spread ≥ 150 mV or $V_{\text{cell}} \leq 2.80$ V), the system forces an immediate **CRITICAL override with 100% confidence**.
- Non-Blocking Asynchronous Mailer:** Outbound notifications are offloaded into an in-memory queue (queue.Queue) handled by background daemon threads, ensuring telemetry ingestion latency remains strictly under 10 ms.
- Local Direct SMTP Routing:** Dispatches alert emails directly from edge hardware to smtp.gmail.com:465 over SSL, completely bypassing cloud datacenter IP blocks. Escalations from WARNING to CRITICAL trigger immediate bypass delivery without cooldown delays.
- Automated Forensics Snapshot:** Outbound emails include active driving mode, exact triggering parameters, pack voltage/current/SOC, all 8 cell voltages, and an **attached CSV containing the last 10 telemetry rows** leading to the event.

6. REAL-TIME DASHBOARD & DEMONSTRATION ORCHESTRATION

- Live Web Interface (live_dashboard_v3.html):** Renders real-time status banners (NORMAL / WARNING / CRITICAL), an 8-cell voltage grid, dynamic charge/discharge mode badges, and a 50-event historical alert table with resolution timestamps.
- Acoustic Alarm Engine:** Employs Web Audio API synthesized audio to emit warning beeps during Stage 2 risks and persistent critical alarm sirens during Stage 3 faults until acknowledged.
- Interactive Fault Replay Launcher (run_dashboard_demo.py):** Features contiguous labeled replay datasets showcasing the full 3-phase lifecycle (Normal Baseline $\rightarrow$ Developing Risk $\rightarrow$ Critical Fault) across all fault classes, plus a 595-row Mixed Full-Cycle (~ 5 min).

7. QUANTITATIVE VALIDATION & KEY ENGINEERING RESULTS

- Early Fault Detection:** Predicts impending weak cell failure and thermal runaway risk **30 to 90 seconds prior** to conventional hardware threshold cutoffs.
- Sub-Second Response Latency:** Detects and isolates cell dropouts, balance wire disconnections, and overcurrent spikes in **< 500 ms**.
- High Predictive Accuracy:** Achieved **98.5%+ classification accuracy** across dynamic drive cycles while maintaining zero false-negative critical escapes.